

INSTALL PYTHON + VS CODE + JUPYTER (.ipynb) + VENV ON WINDOWS 11

STEP 1: INSTALL PYTHON

Go to:

[Python Downloads](#)

Download the latest Python version.

During installation:

IMPORTANT:

✓ Check **"Add Python to PATH"**

Click:

Install Now

After installation, open **Command Prompt** and run:

```
python --version
```

```
pip --version
```

If both commands show version numbers, Python is installed correctly.

STEP 2: INSTALL VS CODE

Go to:

[Visual Studio Code](#)

Download and install VS Code.

During installation, recommended options:

✓ Add to PATH

✓ Open with Code

STEP 3: INSTALL PYTHON EXTENSION IN VS CODE

Open VS Code.

Open Extensions:

```
Ctrl + Shift + X
```

Search:

```
Python
```

Install:

Python (by Microsoft)

STEP 4: INSTALL JUPYTER EXTENSION

In Extensions search:

```
Jupyter
```

Install:

Jupyter (by Microsoft)

This enables `.ipynb` notebook support.

STEP 5: CREATE PROJECT FOLDER

Example folder:

```
D:\PythonProjects\myproject
```

Open this folder in VS Code:

```
File → Open Folder
```

STEP 6: CREATE VIRTUAL ENVIRONMENT (venv)

Open terminal in VS Code:

```
Terminal → New Terminal
```

Run:

```
python -m venv .venv
```

This creates a virtual environment folder named:

```
.venv
```

STEP 7: ACTIVATE VENV

If using PowerShell:

```
.venv\Scripts\Activate.ps1
```

If using Command Prompt:

```
.venv\Scripts\activate.bat
```

After activation you will see:

```
(.venv)
```

at the beginning of the terminal.

STEP 8: FIX POWERSHELL ERROR (ONLY IF NEEDED)

If activation is blocked:

Open PowerShell as Administrator.

Run:

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Press:

```
Y
```

Try activating venv again.

STEP 9: INSTALL JUPYTER INSIDE VENV

With venv activated, run:

```
pip install notebook ipykernel
```

Optional useful libraries:

```
pip install numpy pandas matplotlib
```

STEP 10: SELECT PYTHON INTERPRETER

Press:

```
Ctrl + Shift + P
```

Search:

```
Python: Select Interpreter
```

Choose:

```
.venv
```

STEP 11: CREATE JUPYTER NOTEBOOK

Create a new file:

```
test.ipynb
```

VS Code will open notebook mode automatically.

Select kernel/interpreter at the top-right:

```
Python (.venv)
```

STEP 12: TEST NOTEBOOK

Run this code:

```
print("Hello Jupyter")
```

If output appears, setup is complete.

USEFUL COMMANDS

Install package

```
pip install requests
```

Save installed packages

```
pip freeze > requirements.txt
```

Install packages from requirements file

```
pip install -r requirements.txt
```

RECOMMENDED PROJECT STRUCTURE

```
myproject/  
|  
├─ .venv/  
├─ notebooks/  
├─ src/  
├─ requirements.txt  
└─ README.md
```

OFFICIAL LINKS

Python

[Python Official Website](#)

VS Code

[VS Code Official Website](#)

Python Documentation

[Python Documentation](#)